

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	H	$W_{\mu\nu}$	$F B_{\mu\nu}$	
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)B_{\mu_1\mu_2}^2$ $-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)B_{\mu_1\mu_2}^2$	$+\frac{\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H1})^2}{2\epsilon}$ $+\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H1})^2$ $+\frac{5\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H1})^2}{6\epsilon}$ $+\frac{5}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H1})^2$ $+\frac{\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H2})^2}{2\epsilon}$ $+\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H2})^2$ $+\frac{5\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H2})^2}{6\epsilon}$ $+\frac{5}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H2})^2$	$+\frac{\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H1})^2}{2\epsilon}$ $+\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H1})^2$ $+\frac{5\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H1})^2}{6\epsilon}$ $+\frac{5}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H1})^2$ $+\frac{\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H2})^2}{2\epsilon}$ $+\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H2})^2$ $+\frac{5\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H2})^2}{6\epsilon}$ $+\frac{5}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H2})^2$	$-\frac{\hbar B_{\mu_1\mu_2}^2}{12\epsilon}$ $+C_{H2}H_{a_1}H_{a_1}^\dagger$ $-H_{a_1}\mu^2H_{a_1}^\dagger$ $+\frac{4\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\epsilon}$ $+\frac{4\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3(M_{S_1}^2-M_{S_2}^2)}$ $+\frac{4\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3(M_{S_1}^2-M_{S_2}^2)}$ $-\frac{4\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3(M_{S_1}^2-M_{S_2}^2)}$ $-\frac{4\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3(M_{S_1}^2-M_{S_2}^2)}$ $+\frac{\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H1}^1\lambda_{H1}^2}{\sqrt{3}\epsilon}$ $+\frac{\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H1}^1\lambda_{H1}^2}{\sqrt{3}}$ $+\frac{\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H2}^1\lambda_{H2}^2}{\sqrt{3}\epsilon}$ $+\frac{\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H2}^1\lambda_{H2}^2}{\sqrt{3}}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{12\epsilon}$ $-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)(W_{\mu_1\mu_2}^{A_1})^2$ $-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)(W_{\mu_1\mu_2}^{A_1})^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	H	$W_{\mu\nu}$	$F B_{\mu\nu}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)B_{\mu_1\mu_2}^2$ $-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)B_{\mu_1\mu_2}^2$	$+\frac{\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H1})^2}{2\epsilon}$ $+\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H1})^2$ $+\frac{5\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H1})^2}{6\epsilon}$ $+\frac{5}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H1})^2$ $+\frac{\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H2})^2}{2\epsilon}$ $+\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H2})^2$ $+\frac{5\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H2})^2}{6\epsilon}$ $+\frac{5}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H2})^2$	$+\frac{C_{H2}H_{a_1}H_{a_1}^\dagger}{-H_{a_1}\mu^2H_{a_1}^\dagger}$ $+\frac{4\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\epsilon}$ $+\frac{4\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3(M_{S_1}^2-M_{S_2}^2)}$ $+\frac{4\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3(M_{S_1}^2-M_{S_2}^2)}$ $-\frac{4\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3(M_{S_1}^2-M_{S_2}^2)}$ $-\frac{4\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3(M_{S_1}^2-M_{S_2}^2)}$ $+\frac{\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H1}^1\lambda_{H1}^2}{\sqrt{3}\epsilon}$ $+\frac{\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H1}^1\lambda_{H1}^2}{\sqrt{3}}$ $+\frac{\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H2}^1\lambda_{H2}^2}{\sqrt{3}\epsilon}$ $+\frac{\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H2}^1\lambda_{H2}^2}{\sqrt{3}}$	$-\frac{13\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{12\epsilon}$ $-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$ $-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$
									$+\frac{\hbar H_{a_3}H_{a_4}H_{a_1}^\dagger H_{a_2}^\dagger\mathcal{I}_{H1,a_1,a_3}^1\mathcal{I}_{H1,a_2,a_4}^1(\lambda_{H1})^2}{2\epsilon}$ $+\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)H_{a_3}H_{a_4}H_{a_1}^\dagger H_{a_2}^\dagger\mathcal{I}_{H1,a_1,a_3}^1\mathcal{I}_{H1,a_2,a_4}^1(\lambda_{H1})^2$ $+\frac{3\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H2})^2}{2\epsilon}$ $+\frac{3}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H2})^2$ $+\frac{\text{CH}21H_{a_1}H_{a_1}^\dagger}{-H_{a_1}\mu^2H_{a_1}^\dagger}$ $+\frac{2\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon}$ $+\frac{2\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}(M_{S_1}^2-M_{S_2}^2)}$ $+\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}(M_{S_1}^2-M_{S_2}^2)}$ $-\frac{2\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}(M_{S_1}^2-M_{S_2}^2)}$ $-\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}(M_{S_1}^2-M_{S_2}^2)}$ $+\hbar C_{H2}H_{a_2}H_{a_1}^\dagger\mathcal{I}_{H1,a_1,a_2}^1$	$-\frac{2\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{3\epsilon}$ $-\frac{1}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$-\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)B_{\mu_1\mu_2}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$ $-\frac{1}{4}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)B_{\mu_1\mu_2}^2$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	H	$W_{\mu\nu}$	$F B_{\mu\nu}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$ $-\frac{1}{4}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S1}^2}\right)B_{\mu_1\mu_2}^2$	$+\frac{3\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H1})^2}{2\epsilon}$ $+\frac{3}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H1})^2$ $+\frac{\hbar H_{a_3}H_{a_4}H_{a_1}^\dagger H_{a_2}^\dagger\mathcal{I}_{H2,a_1,a_3}\mathcal{I}_{H2,a_2,a_4}(\lambda_{H2})^2}{2\epsilon}$ $+\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S2}^2}\right)H_{a_3}H_{a_4}H_{a_1}^\dagger H_{a_2}^\dagger\mathcal{I}_{H2,a_1,a_3}\mathcal{I}_{H2,a_2,a_4}(\lambda_{H2})^2$ $+C_{H2}H_{a_1}H_{a_1}^\dagger$ $-H_{a_1}\mu^2H_{a_1}^\dagger$ $+\frac{2\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon}$ $+\frac{2\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}(M_{S1}^2-M_{S2}^2)}$ $+\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_{S1}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}(M_{S1}^2-M_{S2}^2)}$ $-\frac{2\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}(M_{S1}^2-M_{S2}^2)}$ $-\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_{S2}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}(M_{S1}^2-M_{S2}^2)}$ $+CH21\hbar H_{a_2}H_{a_1}^\dagger\mathcal{I}_{H2,a_1,a_2}$	$-\frac{2\hbar\left(W_{\mu_1\mu_2}^{A1}\right)^2}{3\epsilon}$ $-\frac{1}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S1}^2}\right)\left(W_{\mu_1\mu_2}^{A1}\right)^2$	$-\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)B_{\mu_1\mu_2}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$ $-\frac{1}{4}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S2}^2}\right)B_{\mu_1\mu_2}^2$	$+\frac{3\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H1})^2}{2\epsilon}$ $+\frac{3}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H1})^2$ $+\frac{3\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H2})^2}{2\epsilon}$ $+\frac{3}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H2})^2$ $+C_{H2}H_{a_1}H_{a_1}^\dagger$ $-H_{a_1}\mu^2H_{a_1}^\dagger$ $+\frac{2\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon}$ $+\frac{2\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2-M_{S2}^2}$ $+\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_{S1}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2-M_{S2}^2}$ $-\frac{2\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2-M_{S2}^2}$ $-\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_{S2}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2-M_{S2}^2}$	$-\frac{5\hbar\left(W_{\mu_1\mu_2}^{A1}\right)^2}{6\epsilon}$ $-\frac{1}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S1}^2}\right)\left(W_{\mu_1\mu_2}^{A1}\right)^2$ $-\frac{1}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S2}^2}\right)\left(W_{\mu_1\mu_2}^{A1}\right)^2$	$-\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)B_{\mu_1\mu_2}^2$

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F\ H$	$F\ W_{\mu\nu}$	$H\ e$	$F\ H\ e$
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